

sitivity, choose feedback accordingly or scaling factor, likewise. Operating temperature range of IC4 is 0°C to

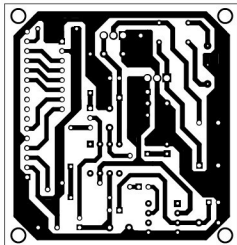


Fig. 4: Actual-size PCB layout of the circuit for interfacing a thermocouple with Arduino

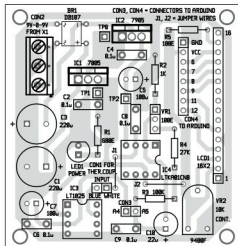


Fig. 5: Component layout of the PCB

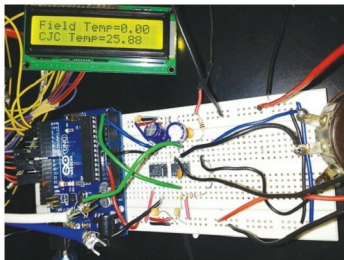


Fig. 6: Prototype on breadboard

70°C. Pin 3 (+IN) of IC4 is connected to the other terminal of CON1. Pin 4 (V-) of IC4 is provided with -5V.

Pin 2 (-IN) of IC4 is connected to potentiometer VR1, which is used to set the amplifier gain.

Here, we have taken an amplifier gain of about 123. By varying VR1, you can change the amplifier gain. Output pin 6 of IC4 is connected to the analogue pin (A5) of Board1.

Arduino UNO (Board1) is a microcontroller (MCU) board based on ATmega328. It has 13 digital pins and six analogue input pins. Here, we are getting analogue inputs from IC3 and IC4. These analogue inputs are converted into digital outputs, which are displayed on the LCD (LCD1) that is connected to digital pin 6 through digital pin 9 of Board1, operating in 4-bit mode.

Software

Circuit operation is done using the software program loaded into the internal memory of Arduino UNO. The program implements all required functionalities including handling user inputs through the thermocouple, converting analogue readings to digital and displaying output on the LCD. The program is written in Arduino programming language sketch. Arduino IDE is used to compile and upload the program.

Construction and testing

An actual-size, single-side PCB layout of the circuit for interfacing a thermocouple with Arduino is shown in Fig. 4 and its component layout is

TABLE II
Test Points

Test point	Details
TP0	0V, GND
TP1	+5V
TP2	-5V

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

shown in Fig. 5. Mount all components on the PCB. It is recommended to use IC bases for IC3 and IC4.

Use a USB A-B cable to upload the compiled sketch (software) from the PC to MCU through Arduino IDE. Unplug the USB cable from the PC and connect Arduino board to CON4 in the PCB.

The operation is as follows:

1. Power on Arduino UNO board. You can view the LCD displays: Field Temp in the first row and CJC Temp in second as shown in Fig. 6. Here, CJC value, which is 25.88, is the room temperature in degree centigrade.

2. Vary 10-kilo-ohm preset VR2 to adjust the contrast of the LCD if the text is not clearly visible. For any other problem, you may verify the voltages at the test points given in Table II.

3. Dip the thermocouple in hot water. When the temperature across the thermocouple is changed, values of temperature on the LCD will also change. You will observe that Field Temp value on the LCD will start going up from 0°C. That means, analogue values from the thermocouple received in the Arduino board are converted into digital and corresponding values of the temperature are displayed on the LCD. ●

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